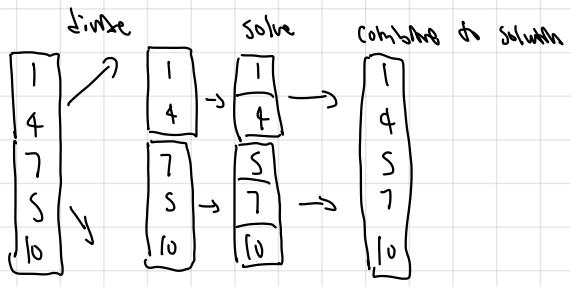
lecture 3 : merge sort
divide and conquer

steps
1. divide input
2. solve problem in subarray
3. combine to solve problem



solve problem with recursion

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merge sort
- divide list into 2 smaller lists

note: don't create new lists to save space
receive lower and upper bound indices
[l, h)

Parameters
mergesort([10,5,7,4,1], 0, 5)

for a recursive algorithm, define the base and recursive case

Definition:

```
merge_sort(A, l, h):
    if(h - l ≤ 1):
        return

    m = l + (h - l)//2
    merge_sort(A, l, m)
    merge_sort(A, m, h)

    merge(A, l, m, h):
```

[3, 4, 5] [1, 2, 7]

```
merge(A, l, m, h):
    T = [scratch space of size >= N]
    k, i, j = 0, l, m

    while i<m and j<h:
        if A[i] <= A[j]:
            T[k++] = A[i++]
        else:
            T[k++] = A[j++]

    while i<m:
        T[k++] = A[i++]
    while j<h:
        T[k++] = A[j++]

    for i in 0, 1, ..., (h-l)-1:
        A[l+i] = T[i]
```

---

merge sort example

A = [4,3,2,1] N = 4
0 1 2 3

after merge:
MS(A,0,4) A = [1,2,3,4]

after merge
A = [3,4,2,1]
MS(A,0,2)

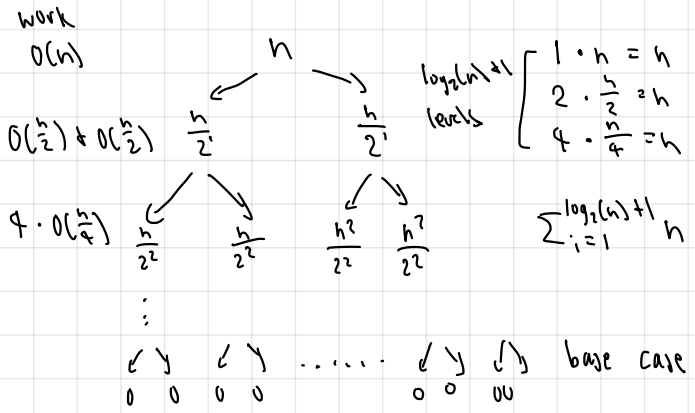
after merge
A = [3,4,1,2]
MS(A,2,4)

MS(A,0,1) MS(A,1,2) MS(A,2,3) MS(A,3,4)

levels: $\log_2(n)+1$
$= \log_2(4)+1 = 2+1 = 3$

assume n is a power of 2

work
$O(n)$



$$\sum_{i=1}^{\log_2(n)+1} n = n \cdot \sum_{i=1}^{\log_2(n)+1} 1$$
$$= n(\log_2(n)+1 - 1 + 1)$$
$$= n \cdot \log_2(n+1)$$

proof:

Define $T(n)$ to be the runtime of MS

base $T(1) \leq C$

$n = 2^k$

recursive $T(2^k) \leq \underbrace{T(2^{k-1}) + T(2^{k-1})}_{\text{merge sort calls}} + \underbrace{C \cdot 2^k}_{\text{merge}}$

$= 2T(2^{k-1}) + C \cdot 2^k$

use $T(2^{k-1}) = T(2^{k-1-1}) + T(2^{k-1-1}) + C \cdot 2^{k-1}$

$= 2[T(2^{k-2}) + T(2^{k-2}) + C \cdot 2^{k-1}] + C \cdot 2^k$

$= 2[2T(2^{k-2}) + C \cdot 2^{k-1}] + C \cdot 2^k$

$= 4T(2^{k-2}) + 2 \cdot C \cdot 2^k$

$= 2^2 T(2^{k-2}) + 2^1 \cdot C \cdot 2^k$

$\vdots$

$\leq 2^k T(2^0) + k(C2^k)$

$\leq 2^k O(1) + kC2^k$

$\leq O(2^k k)$

$\leq O(2^{\log_2(n)} \log_2(n))$

$\leq O(n \cdot \log_2(n))$

recall $n = 2^k$

$\log_2(n) = \log_2(2^k)$

$\log_2(n) = k \cdot \log_2(2)$

$k = \log_2(n)$

Induction:

example: $a_n = 2a_{n-1}$

$a_0 = 1$

prove: $a_n = 2^n$